

Using Q_w , the between-study variance, η^2 , is:

$$\eta^2 = \frac{Q_w - (n-1)}{\sum 1/v_i - \frac{\sum 1/v_i^2}{\sum 1/v_i}}$$

It can be shown that the denominator is always positive. Therefore, if the numerator is negative (i.e., if $Q_w < n-1$), then η^2 is a negative number, and it is not possible to calculate a random effects estimate. In this case, however, the small value of Q_w would presumably have led to accepting the null hypothesis described above, and the meta-analysis would be based on the fixed effects model. The remaining discussion therefore assumes that η^2 is positive.

Given a value for η^2 , the random effects estimate is calculated in almost the same way as the fixed effects estimate. However, the weights now incorporate both the within-study variance (v_i) and the between-study variance (η^2). Whereas the weights implied by the fixed effects model used only v_i , the within-study variance, the weights implied by the random effects model use $v_i + \eta^2$.

Let $v_i^* = v_i + \eta^2$. Then:

$$S^* = \sum \frac{1}{v_i^*},$$

$$w_i^* = \frac{1/v_i^*}{S^*}$$

The estimate produced by pooling based on the random effects model, then, is just a weighted average of the estimates from the studies being considered, with the weights as defined above. That is:

$$\beta_{rand} = \sum w_i^* \times \beta_i$$

The variance associated with this random effects pooled estimate is, as it was for the fixed effects pooled estimate, the inverse of the sum of the inverse variances:

$$v_{rand} = \frac{1}{\sum 1/v_i^*}$$

The weighting scheme used in a pooling based on the random effects model is basically the same as that used if a fixed effects model is assumed, but the variances used in the calculations are different. This is because a fixed effects model assumes that the variability among the estimates from different studies is due only to sampling error (i.e.,

each study is thought of as representing just another sample from the same underlying population), while the random effects model assumes that there is not only sampling error associated with each study, but that there is also between-study variability -- each study is estimating a different underlying β . Therefore, the sum of the within- study variance and the between-study variance yields an overall variance estimate.

Fixed Effects and Random / Fixed Effects Weighting to Pool Incidence Change Distributions and Dollar Benefit Distributions

Weights can be derived for pooling incidence changes predicted by different studies, using either the fixed effects or the fixed / random effects model, in a way that is analogous to the derivation of weights for pooling the β 's in the C-R functions. As described above, BenMAP generates a Latin Hypercube representation of the distribution of incidence change corresponding to each health impact function selected. The means of those study-specific Latin Hypercube distributions of incidence change are used in exactly the same way as the reported β 's are used in the calculation of fixed effects and random effects weights described above. The variances of incidence change are used in the same way as the variances of the β 's. The formulas above for calculating fixed effects weights, for testing the fixed effects hypothesis, and for calculating random effects weights can all be used by substituting the mean incidence change for the i th health impact function for β_i and the variance of incidence change for the i th health impact function for v_i .

Similarly, weights can be derived for dollar benefit distributions. As described above, BenMAP generates a Latin Hypercube representation of the distribution of dollar benefits. The means of those Latin Hypercube distributions are used in exactly the same way as the reported β 's are used in the calculation of fixed effects and random effects weights described above. The variances of dollar benefits are used in the same way as the variances of the β 's. The formulas above for calculating fixed effects weights, for testing the fixed effects hypothesis, and for calculating random effects weights can all be used by substituting the mean dollar benefit change for the i th valuation for β_i and the variance of dollar benefits for the i th valuation for v_i .

BenMAP always derives Fixed Effects and Random / Fixed Effects weights using nationally aggregated results, and uses those weights for pooling at each grid cell (or county, etc. if you choose to aggregate results prior to pooling). This is done because BenMAP does not include any regionally based uncertainty - that is, all uncertainty is at the national level in BenMAP, and all regional differences (population, for example) are treated as certain.

K.2.2 Mechanics of Pooling in BenMAP

Once weights are generated for each input distribution, BenMAP has three options for using these weights to combine the input distributions into a single new distribution. These options are referred to as Advanced Pooling Methods.

Round Weights to Two Digits

This is BenMAP's default Advanced Pooling Method, and is always the method used when Subjective Weights are used. The first step is converting the weights to two digit integers by multiplying them by 100 and rounding to the nearest integer. If all the integral weights thus generated are divisible by the smallest weight, they are each divided by that smallest weight. For example, if the original weights were 0.1, 0.2, 0.3, and 0.4, the resulting integral weights would be 10/10, 20/10, 30/10, and 40/10 (or 1, 2, 3, and 4).

BenMAP then creates a new distribution by sampling each entire input distribution according to its weight. That is, in the above example the first distribution would be sampled once, the second distribution twice, and so forth. The advantage of sampling whole distributions is that it preserves the characteristics (i.e., the moments - the mean, the variance, etc.) of the underlying distributions. Assuming n latin hypercube points, the resulting distribution will contain a maximum of $100 * n$ values, which are then sorted low to high and binned down to n values, which will represent the new, pooled distribution.

Round Weights to Three Digits

This Advanced Pooling Method is essentially the same as rounding weights to two digits, except that the weights are converted to three digit integers, and so forth. That is, the weights are multiplied by 1000 and rounded to the nearest integer. Again, if all the integral weights thus generated are divisible by the smallest weight, they are each divided by that smallest weight. Assuming n Latin Hypercube points, the resulting distribution with this Advanced Pooling Method can contain a maximum of $1000 * n$ values, which are sorted low to high and binned down to n values, which represent the new, pooled distribution.

Exact Weights for Monte Carlo

This Advanced Pooling Method uses a Monte Carlo method to combine the input distributions. Using this method, on each of many iterations, (1) an input distribution is selected (with the probability of selection equal to the weight assigned to the distribution), and (2) a value is randomly drawn from that distribution. Values chosen in this way are placed into a temporary pooled distribution, which will have one point per iteration of the Monte Carlo method. The number of iterations is specified by the user, and defaults to 5,000. After the temporary distribution is fully generated, it is sorted low to high and binned down to n values (where n is the number of Latin Hypercube Points chosen for the analysis).

K.2.3 Summing Distributions

Sometimes rather than pooling distributions we want to add them. For example, some studies have estimated a health impact function for hospital admissions for COPD and another health impact function for hospital admissions for pneumonia. From each of these health impact functions, BenMAP can derive the corresponding distributions for

incidence change. Hospital admissions for COPD and pneumonia are two of the most important components of respiratory hospital admissions, and we may want to estimate the number of cases of “respiratory hospital admissions,” as characterized by being either COPD or pneumonia. To do this we would add the two distributions.

Summing across distributions can be done in one of two ways: We can assume the two distributions are independent of each other or dependent. Which is the more reasonable assumption depends on the particulars of the distributions being summed.

Assuming Independence

This is the Sum (Independent) Pooling Method. To sum two distributions that are independent, on each of many iterations of a Monte Carlo procedure, BenMAP (1) randomly selects a value from the first input distribution, (2) randomly selects a value from the second input distribution, and (3) adds the two values together. To sum N distributions that are independent, BenMAP follows an analogous procedure in which, on each iteration it makes a random selection from each of the input distributions and then adds the results together. When the Monte Carlo procedure is completed, all such generated results are sorted low to high and binned down to the appropriate number of latin hypercube points. The number of iterations is determined by the Monte Carlo Iterations setting.

Assuming Dependence

This is the Sum (Dependent) Pooling Method. Recall that the uncertainty distributions in BenMAP are latin hypercube representations, consisting of N percentile points. To sum two distributions assumed to be dependent, BenMAP simply generates a new N point latin hypercube where each point is the sum of the corresponding points from the input latin hypercubes. That is, the first point in the new latin hypercube is the sum of the first points in the two input latin hypercubes, and so forth. To sum n distributions that are assumed to be dependent, BenMAP follows an analogous procedure in which each point in the new latin hypercube is the sum of the corresponding points from each of the input latin hypercubes.

K.2.4 Subtracting Distributions

In some cases, you may want to subtract one or more distribution(s) from another. For example, one study may have estimated a health impact function for minor restricted activity days (MRADs), and another study may have estimated a health impact function for asthma “episodes.” You may want to subtract the change in incidence of asthma episodes from the change in incidence from MRADs before estimating the monetary value of the MRADs, so that the monetary value of asthma episodes avoided will not be included.

Subtracting across distributions can be done in one of two ways: we can assume the two distributions are independent of each other or dependent. Which is the more